MAST20030 Differential Equations
— Problem Sheet 0: Revision —

1. Consider the two sequences of functions

$$f_N(x) = \mathrm{e}^{\frac{x}{N}} \qquad \text{and} \qquad g_N(x) = x^N$$

For each function:

(a) Plot the function for $N = \{1, 2, 5, 10\}$ on the domain $x \in [0, 1]$, using the same vertical axis scale in each case.

(b) Find the (piecewise) limit for the sequence of functions:

$$\lim_{N\to\infty} f_N(x) \qquad \text{and} \qquad \lim_{N\to\infty} g_N(x)$$

for $x \in [0, 1]$.

2. Solve the following problems:

(a) $P'(t) + P(t) = 0$ subject to $P(0) = 1$

(b) $P'(t) - P(t) = 0$ subject to $P(0) = 2$

(c) $P''(t) + P(t) = 0$ subject to $P'(0) = 1$ and $P(0) = 0$

(d) $P''(x) - P(x) = 0$ subject to $P(0) = 0$ and $P(1) = 1$

3. Evaluate the following integrals:

(a) $\displaystyle\int_0^1 t^2 e^{t^3}\, dt$ (b) $\displaystyle\int \sqrt{1 + 4x^2}\, dx$

(c) $\displaystyle\int t^2 \sinh t dt$ (d) $\displaystyle\int \sin^2 2x\, dx$

(e) $\displaystyle\int e^x \cos 3x\, dx$ (f) $\displaystyle\int_0^{\frac{\pi}{2}} \sin^3 t\, dt$

(g) $\displaystyle\int_0^{\pi} t \cos(\omega t)\, dt$ (h) $\displaystyle\int \sin x \sin 4x dx$

4. Evaluate the following limits if they exist:

(a) $\displaystyle\lim_{x\to\infty} \frac{3x + 5}{x - 4}$ (b) $\displaystyle\lim_{x\to\infty} \frac{x^3 + 5x}{2x^3 - x^2 + 4}$

(c) $\displaystyle\lim_{x\to\infty} \left(\sqrt{9x^2 + x} - 3x\right)$ (d) $\displaystyle\lim_{x\to\infty} \cos x$

(e) $\displaystyle\lim_{x\to\infty} \left[\log(x^2 - 1) - \log(x + 1)\right]$ (f) $\displaystyle\lim_{x\to\infty} \left(x^2 e^{-x}\right)$

5. Determine the eigenvalues and eigenvectors of the following matrices:

(a) $\begin{bmatrix} 4 & 2 \\ -3 & -1 \end{bmatrix}$ (b) $\begin{bmatrix} 3 & -2 \\ 2 & -1 \end{bmatrix}$ (c) $\begin{bmatrix} 3 & -2 \\ 17 & -7 \end{bmatrix}$

6. Diagonalise, if possible, the following matrices. In each case, check using matrix multiplication that your diagonalisation is correct.

(a) $\begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}$ (b) $\begin{bmatrix} 3 & 4 \\ -1 & -2 \end{bmatrix}$ (c) $\begin{bmatrix} 3 & 0 \\ 3 & 3 \end{bmatrix}$

7. Express the following rational functions as a sum of real partial fractions:

   (a) $\dfrac{2x^2 - x + 4}{x(x^2 + 4)}$ (b) $\dfrac{4x}{(x-1)^2(x+1)}$ (c) $\dfrac{s^2 + 1}{s^2(s-1)^2}$

8. Solve the following first order differential equations:

   (a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = e^{x-2y}$ (b) $\dfrac{\mathrm{d}x}{\mathrm{d}t} - 6tx = 5t$

   (c) $\dfrac{\mathrm{d}y}{\mathrm{d}t} - 2y = 0, \quad y(0) = 1$

9. Solve the following second order differential equations:

   (a) $y'' - 9y = 0$ (b) $\ddot{x} - 12\dot{x} + 36x = 0$

   (c) $\dfrac{\mathrm{d}^2x}{\mathrm{d}t^2} + 8\dfrac{\mathrm{d}x}{\mathrm{d}t} + 17x = 0$ (d) $\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4y = 0, \quad y(0) = 1, \quad y'(0) = 1$

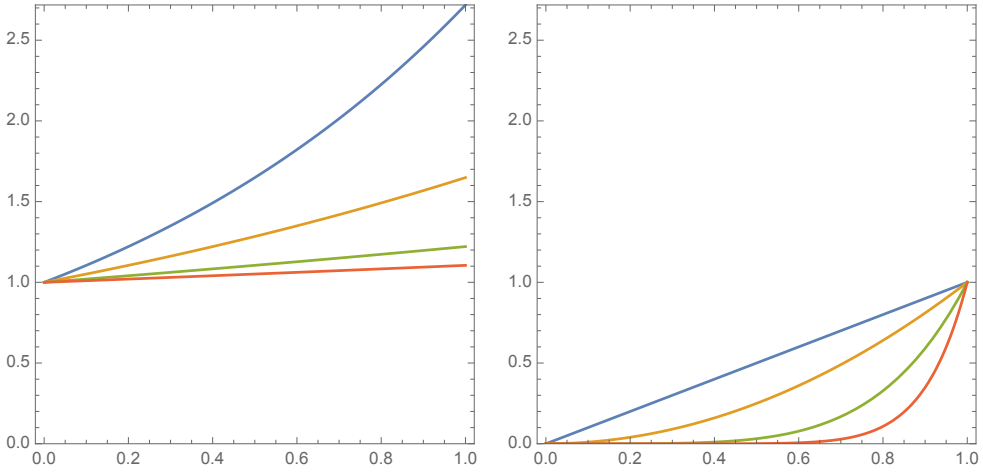
10. Find all first order partial derivatives of the following functions:

    (a) $f(x, y, z) = 3x^2 + 2xy - z^5$ (b) $f(x, y, z) = x^3 \cosh(2y) \sin(5z)$

    (c) $f(x, y, z) = 2xye^z + x^2 \cos(yz)$ (d) $f(x, y, z) = \log(x - y^2 + z^3)$

MAST20030 Differential Equations
— Solutions to Problem Sheet 0: Revision —

1. (a)



(b)

$$\lim_{N\to\infty} f_N(x) = 1 \qquad \text{text} \qquad \lim_{N\to\infty} g_N(x) = \begin{cases} 0 & x < 1 \\ 1 & x = 1 \end{cases}$$

2.

(a) e^{-t}

(b) $2\mathrm{e}^{t}$

(c) $\sin t$

(d) $\dfrac{2\mathrm{e}\sinh t}{\mathrm{e}^2 - 1}$

3.

(a) $\dfrac{1}{3}(\mathrm{e} - 1)$

(b) $\dfrac{1}{4}\left(2x\sqrt{1+4x^2} + \operatorname{arcsinh} 2x\right) + C$

(c) $(2 + t^2)\cosh t - 2t\sinh t + \mathrm{C}$

(d) $\dfrac{x}{2} - \dfrac{1}{8}\sin 4x + C$

(e) $\dfrac{1}{10}\mathrm{e}^{x}(\cos 3x + 3\sin 3x) + C$

(f) $\dfrac{2}{3}$

(g) $\dfrac{\cos\omega\pi + \omega\pi\sin\omega\pi - 1}{\omega^2}$

(h) $\dfrac{1}{6}\sin 3x - \dfrac{1}{10}\sin 5x + C$

4.

(a) 3

(b) $\dfrac{1}{2}$

(c) $\dfrac{1}{6}$

(d) Limit does not exist

(e) Limit does not exist

(f) 0

5.

(a) $\lambda_1 = 1$, $\boldsymbol{v}_1 = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$ $\lambda_2 = 2$, $\boldsymbol{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

(b) $\lambda_1 = 1$, $\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

(c) $\lambda_{1,2} = -2 \pm 3i$, $\boldsymbol{v}_{1,2} = \begin{bmatrix} 5 \pm 3i \\ 17 \end{bmatrix}$

6.

(a) $\begin{bmatrix} 1 & -2 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \cdot \frac{1}{3} \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} -4 & -1 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \cdot \frac{1}{3} \begin{bmatrix} -1 & -1 \\ 1 & 4 \end{bmatrix}$

(c) Not diagonalisable.

7.

(a) $\frac{1}{x} - \frac{1}{x^2+4} + \frac{x}{x^2+4}$ (b) $\frac{1}{x-1} + \frac{2}{(x-1)^2} - \frac{1}{x+1}$ (c) $\frac{2}{s} + \frac{1}{s^2} - \frac{2}{s-1} + \frac{2}{(s-1)^2}$

8.

(a) $\frac{1}{2}\log(2\mathrm{e}^x + C)$ (b) $-\frac{5}{6} + C\mathrm{e}^{3t^2}$

(c) e^{2t}

9. Solve the following second order differential equations:

(a) $C_1\mathrm{e}^{3x} + C_2\mathrm{e}^{-3x}$ (b) $C_1\mathrm{e}^{6t} + C_2 t\mathrm{e}^{6t}$

(c) $\mathrm{e}^{-4t}(C_1\cos t + C_2\sin t)$ (d) $\cos 2t + \frac{1}{2}\sin 2t$

10. Find all first order partial derivatives of the following functions:

(a) $6x + 2y$, $2x$ and $-5z^4$

(b) $3x^2\cosh 2y \sin 5z$, $2x^3\sinh 2y \sin 5z$ and $5x^3\cosh 2y\cos 5z$

(c) $2y\mathrm{e}^z + 2x\cos yz$, $2x\mathrm{e}^z - x^2 z\sin yz$ and $2xy\mathrm{e}^z - x^2 y\sin yz$

(d) $\frac{1}{x-y^2+z^3}$, $-\frac{2y}{x-y^2+z^3}$ and $\frac{3z^2}{x-y^2+z^3}$